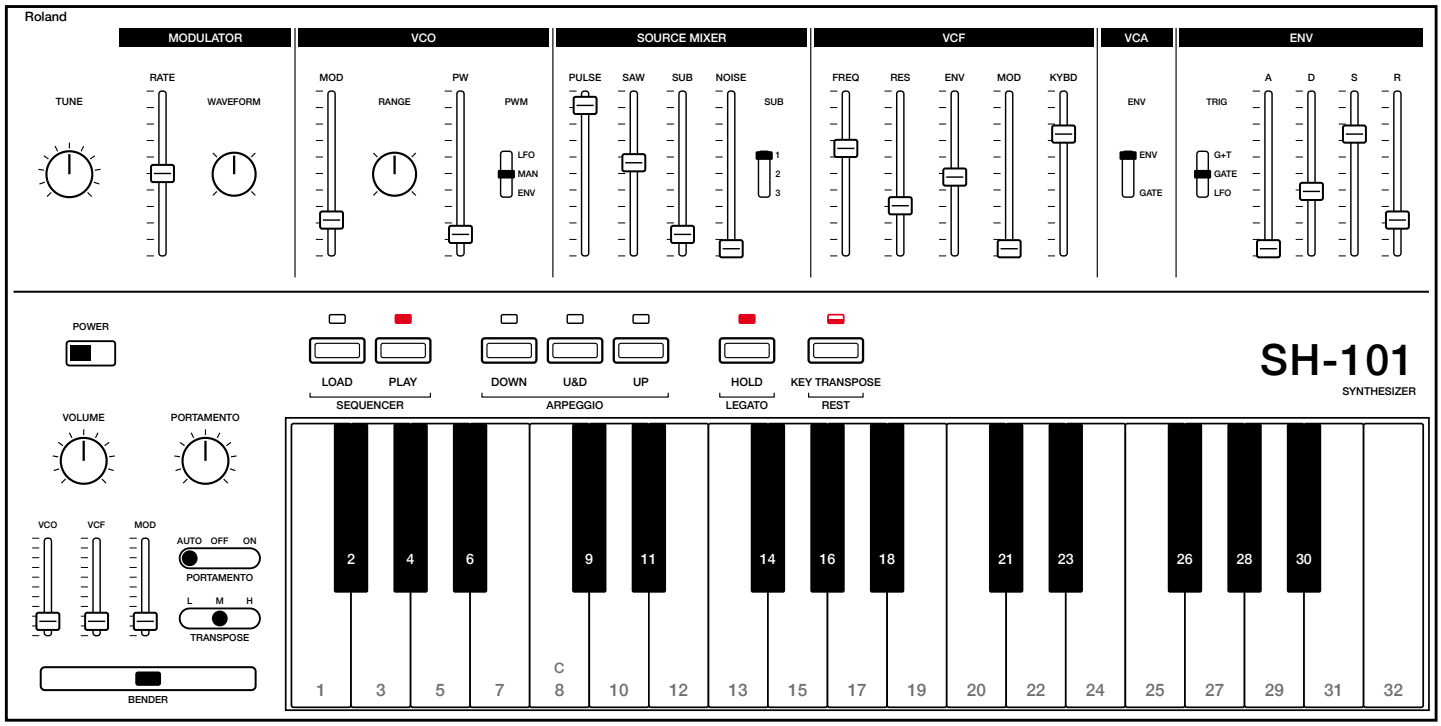
SuperOS-101

SHORTCUT GUIDE

BUILD 2026-09-28 (135021E*) · CUSTOM FIRMWARE FOR THE ROLAND SH-101
AT90USB1286 CPU BOARD · SUPEROS + 80C49 EMULATOR · MIDI ON DIN AND USB-C

Every SuperOS function on one drawing each: the panel, the controls involved, the order to press them, and the LEDs the SH-101 shows while you do it. The LED states are real: the firmware ran each sequence in the host simulator. Find the function, look at the picture, follow the numbered steps.



READING THE DRAWINGS

○ Black ring: press this.
● Red ring and red label: hold it while you do the rest.
- - - Blue dashed box: any key in that group.

LEDS

● lit ◒ blinking ○ off
The box beside each drawing repeats the seven LEDs large. Keys are numbered 1 to 32 from the lowest F; key 8 is the lowest C.

WORDS

A + B: hold A, then press B. Tap: press and release. Playing: the sequencer runs. Clock: the LFO / CLK RATE, or MIDI clock with sync on. The header gives the mode and when it works.

CONTENTS

PLAYING

1 Keyboard 3
2 Hold 3
3 Transpose Switch 4
4 Portamento Auto 4

ARPEGGIO

5 Arpeggio 5
6 Random Arpeggio 5
7 Arpeggio Octave Range 6
8 Arpeggio and Hold 6

SEQUENCER

9 Record a Pattern 7
10 Ties, Rests, Ratchets, Undo 7
11 Play and Stop 8
12 Correct and Add 8
13 Real-Time Record 9
14 Pattern Select 9
15 Song 10
16 Instant Switch 10
17 Play Direction 11
18 Shift 11
19 Loop Length, Live 12

TRANSPOSE AND TUNE

20 Key Transpose 12
21 Transpose Mode 13
22 Pedal as Key Transpose 13
23 Tune 14

CONFIG MENU

24 Enter and Leave 14
25 Firmware: SuperOS or Original 15
26 MIDI Clock Sync 15
27 Clock Division 16
28 Accent, Cutoff, LFO Sync 16
29 Direction and Pedal 17
30 Sync Delay 17

MIDI SYNC

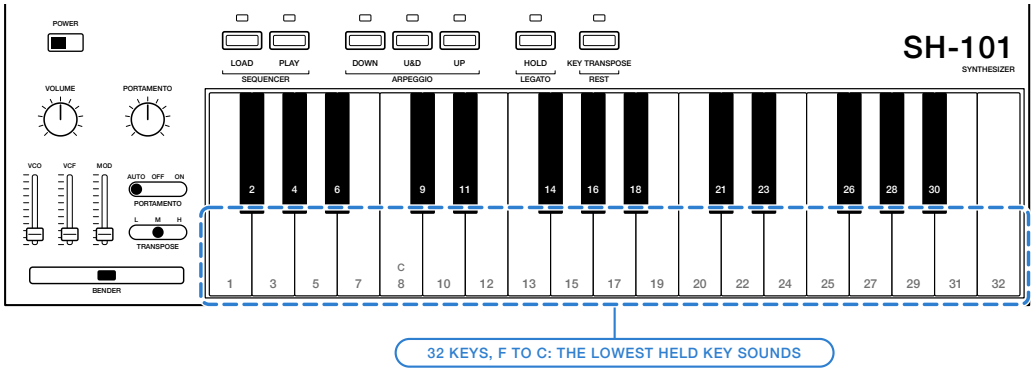
31 Host Transport 18

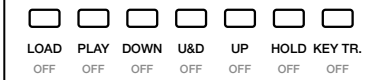
REFERENCE

LEDs, every gesture, MIDI 19

1 KEYBOARD

KEYBOARD | NOTHING LIT





No LED: plain keyboard mode. Keys 8 and 15 held, key 8 (the lower) sounds.

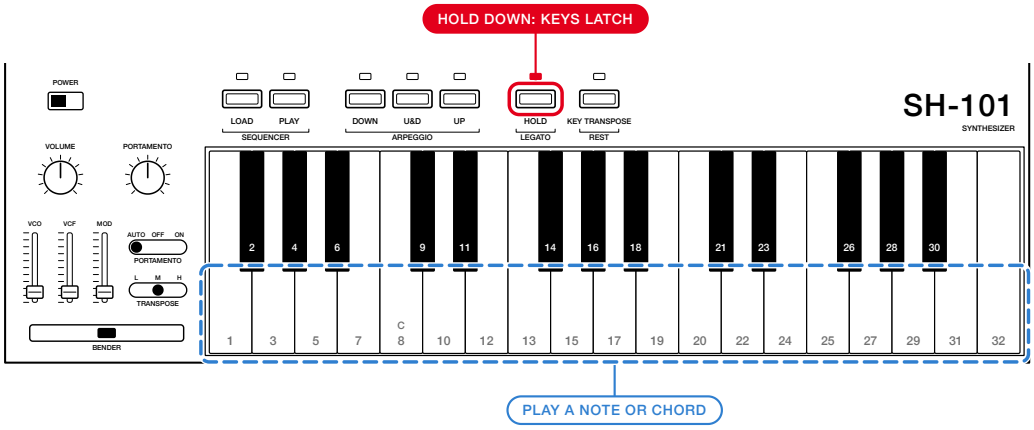
1. With no function LED lit, the keyboard plays the voice directly.
2. Hold two keys: the lower one sounds. Let it go and the other one takes over.
3. Key 1 is the lowest F, key 8 the lowest C, key 32 the top C.

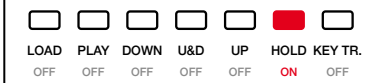
NOTE

With the ENV TRIG switch on GATE+TRIG the priority is the last key pressed instead. MIDI notes 53 to 84 act as keys 1 to 32.

2 HOLD

KEYBOARD OR ARPEGGIO | ANY TIME





HOLD lit while the button (or the pedal) is down: the keys played keep sounding.

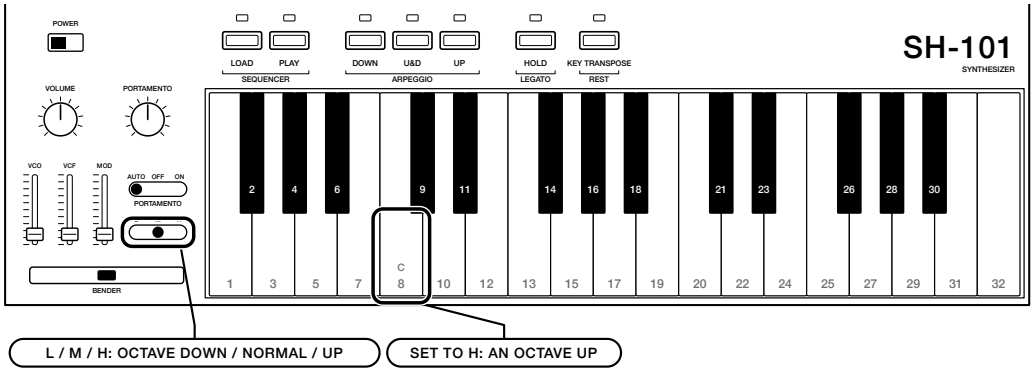
1. Keep HOLD pressed, or press the HOLD pedal.
2. Play keys: they keep sounding after you let go. A new chord replaces the latched one.
3. Release HOLD (or the pedal): the latched keys are released and the LED goes out.

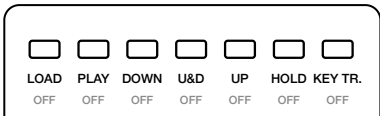
NOTE

The HOLD LED follows the button and the pedal: it is lit only while one of them is down. A pedal on the HOLD jack is the hands-free way to latch.

3 TRANSPOSE SWITCH

KEYBOARD | ANY TIME





No LED. The switch is read with every key.

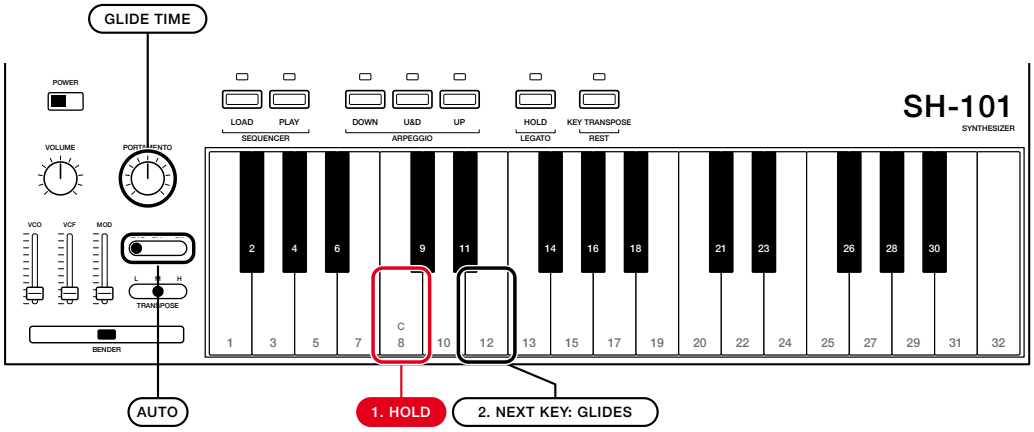
1. Set TRANSPOSE to L, M or H: the keyboard plays an octave down, as marked, or an octave up.
2. Keys you record take the octave the switch is at; a stored pattern keeps it.

NOTE

The TRANSPOSE switch works on keys, the arpeggio and recording, as on the stock SH-101. Tune (below) assumes M.

4 PORTAMENTO AUTO

KEYBOARD, ARPEGGIO, SEQUENCER | SWITCH ON AUTO





No LED. Key 12 pressed while key 8 is held: legato, so it glides.

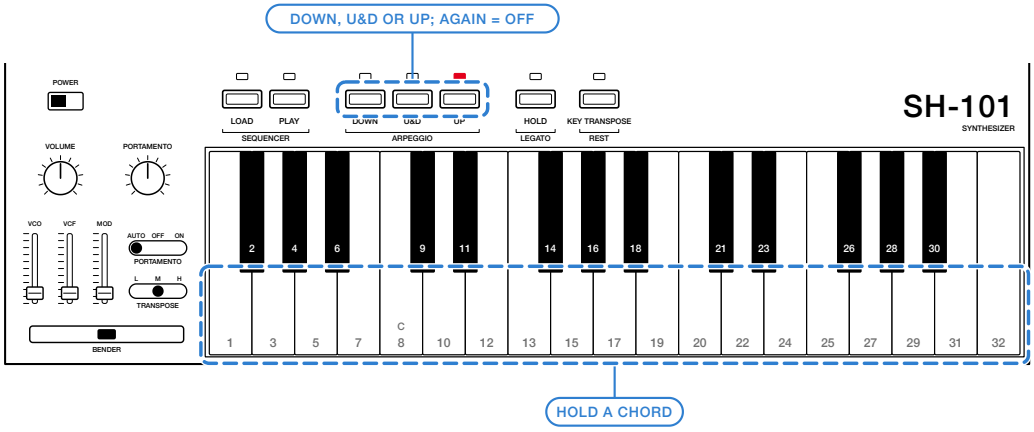
1. Set the PORTAMENTO switch to AUTO and the PORTAMENTO knob to the glide time.
2. Play legato (the next key before you let go of the last): the pitch glides.
3. Play detached: each note starts on its own pitch.

NOTE

The firmware pulses PORTAMENTO OFF on every fresh note, so on AUTO only legato keys and legato (tied) sequencer steps glide. ON and OFF ignore it, as stock.

5 ARPEGGIO

ARPEGGIO | FROM KEYBOARD MODE



LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	OFF	OFF	OFF	ON	OFF	OFF

UP lit: arpeggio up over keys 8, 12, 15.

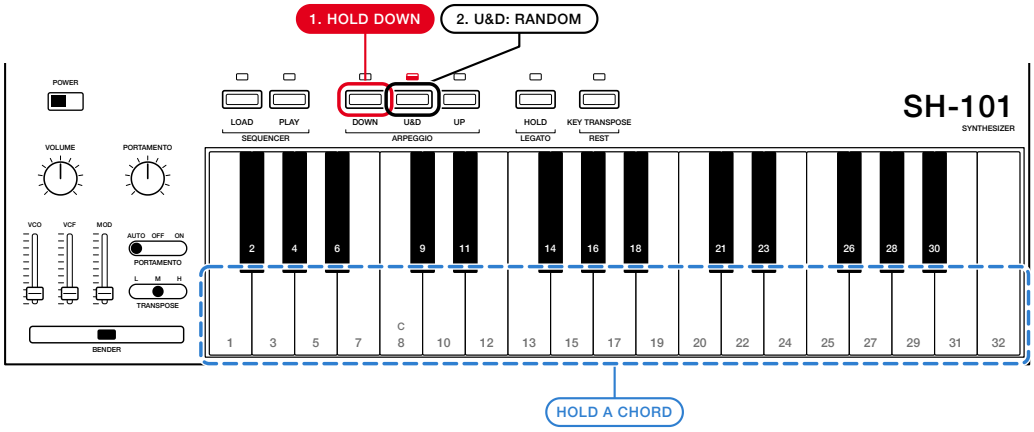
1. Press UP, DOWN or U&D: its LED lights.
2. Hold a chord: the notes play one after another at the LFO / CLK RATE (or the MIDI clock with sync on).
3. Press another direction to change it; press the lit button again to go back to the keyboard.

NOTE

Each arpeggio note retriggers the envelope; CV / GATE OUT and MIDI OUT follow it.

6 RANDOM ARPEGGIO

ARPEGGIO | FROM KEYBOARD OR ARPEGGIO



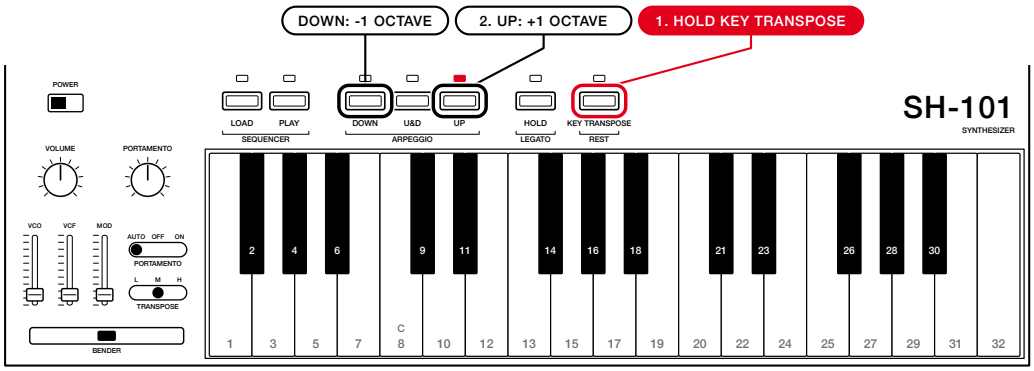
LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	OFF	OFF	BLINK	OFF	OFF	OFF

U&D blinking: random arpeggio.

1. Press DOWN and U&D together: the U&D LED blinks.
2. Held keys play in random order, never the same note twice in a row when there is a choice.
3. DOWN + U&D together again: back to the keyboard.

7 ARPEGGIO OCTAVE RANGE

ARPEGGIO | ARPEGGIO ON



LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	OFF	OFF	OFF	ON	OFF	OFF

UP still lit: the range changed, the mode did not. No LED shows the range.

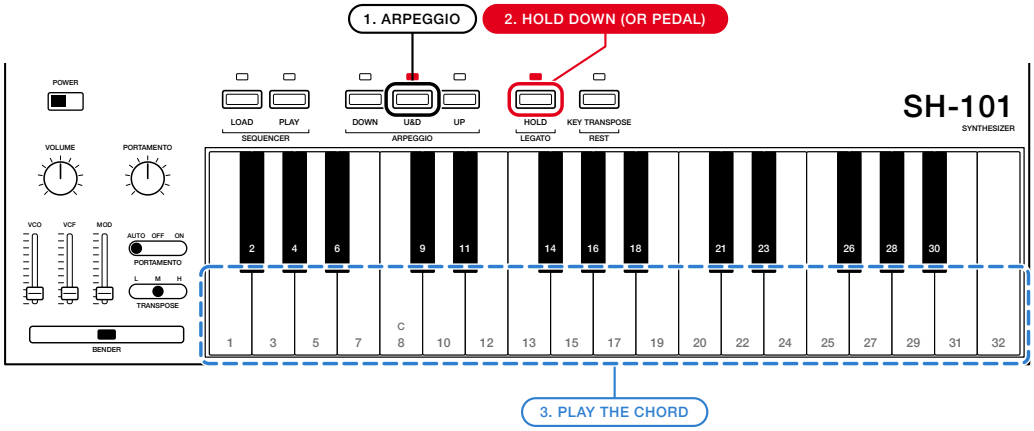
1. With an arpeggio on, hold KEY TRANSPOSE.
2. UP adds an octave, DOWN takes one away: 1 to 3 octaves.
3. UP climbs octave by octave, DOWN starts at the top, U&D bounces across the whole span, random picks an octave too.

NOTE

The range is kept across power-off. It needs an arpeggio running: in keyboard mode KEY TRANSPOSE + UP does nothing to the range.

8 ARPEGGIO AND HOLD

ARPEGGIO | ARPEGGIO ON



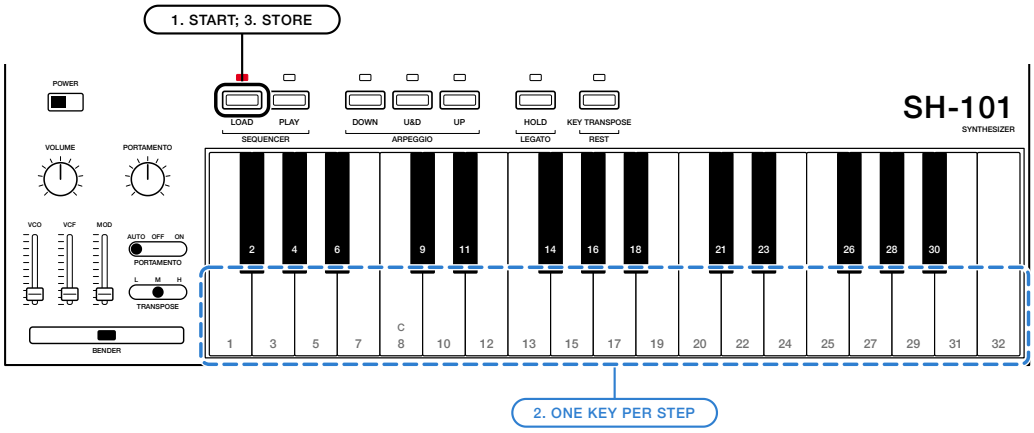
LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	OFF	OFF	ON	OFF	ON	OFF

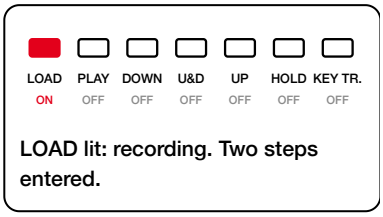
U&D and HOLD lit: the chord keeps arpeggiating after the keys are released.

1. Pick an arpeggio direction.
2. Keep HOLD down, or press the pedal, and play the chord: it keeps running with the keys released.
3. A new chord replaces it. In this state transpose mode (Transpose chapter) moves it with single keys.

9 RECORD A PATTERN

LOAD | FROM STOP





LOAD lit: recording. Two steps entered.

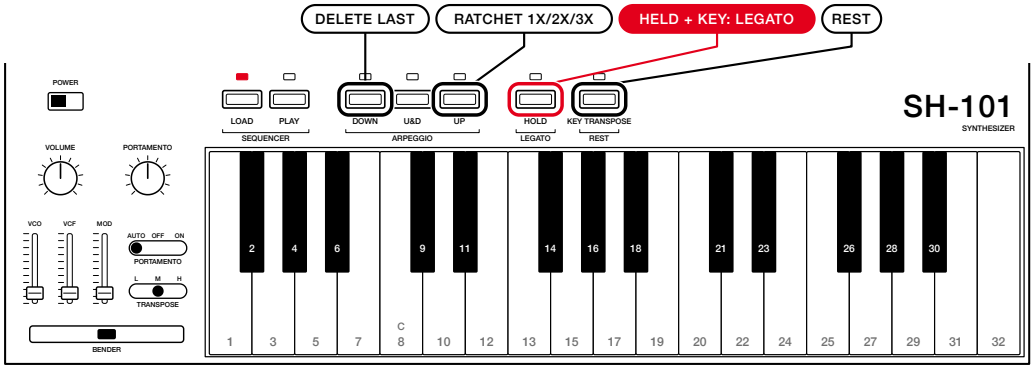
1. Stopped, press LOAD: the LED lights and the current pattern is cleared.
2. Play the notes, one key per step. Timing does not matter: every key is one step.
3. Press LOAD again: the pattern is stored and the LED goes out.

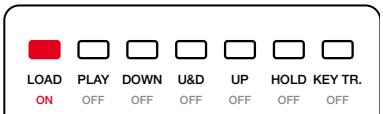
NOTE

Up to 128 steps, 8 patterns, kept in flash across power-off. PLAY + key 1-8 picks the pattern to record into (Pattern Select).

10 TIES, RESTS, RATCHETS, UNDO

LOAD | RECORDING





LOAD lit. Recorded: key 8, key 10 tied, a rest, key 12 (then ratcheted and taken back).

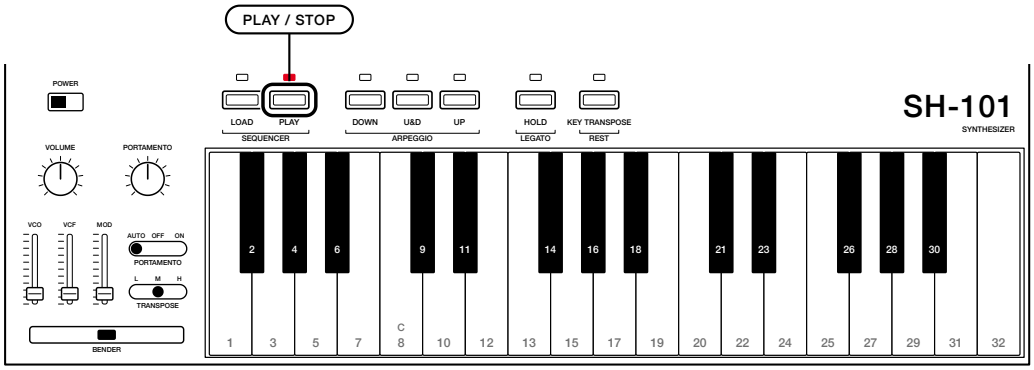
1. Hold HOLD while pressing a key: that step is legato (no retrigger, glides on AUTO).
2. KEY TRANSPOSE enters a rest.
3. UP cycles the last step's ratchet: 1x, 2x, 3x hits inside the step.
4. DOWN takes the last step back; press it again to go further back.

NOTE

The stock panel prints LEGATO under HOLD and REST under KEY TRANSPOSE for this.

11 PLAY AND STOP

SEQUENCER PLAY | KEYBOARD MODE





PLAY lit: the pattern runs at the LFO / CLK RATE.

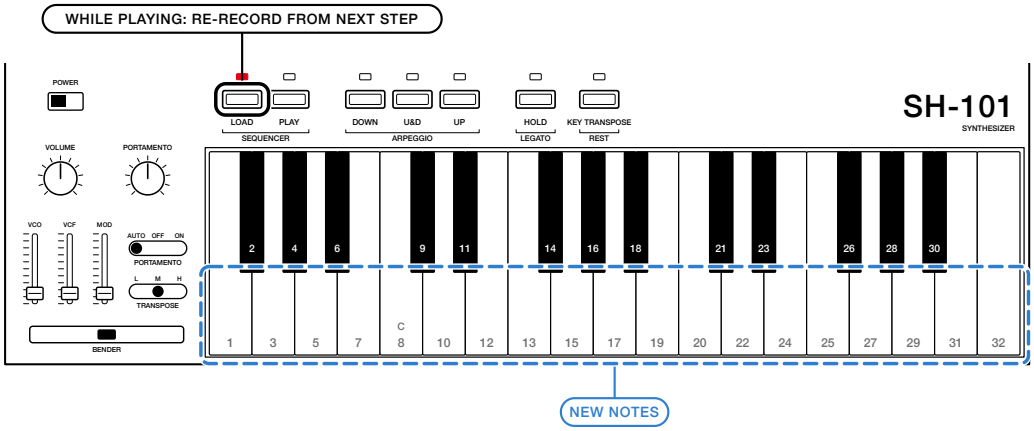
1. Press PLAY: the LED lights and step 1 sounds at once, the next steps at the LFO / CLK RATE (or the MIDI clock with sync on).
2. Press PLAY again to stop.

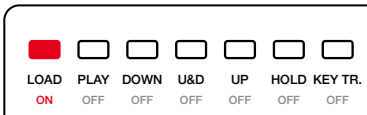
NOTE

With MIDI sync on, PLAY only arms: the first step waits for the clock (MIDI Sync).

12 CORRECT AND ADD

LOAD | PLAYING





LOAD lit: recording again from the step after the one that was sounding.

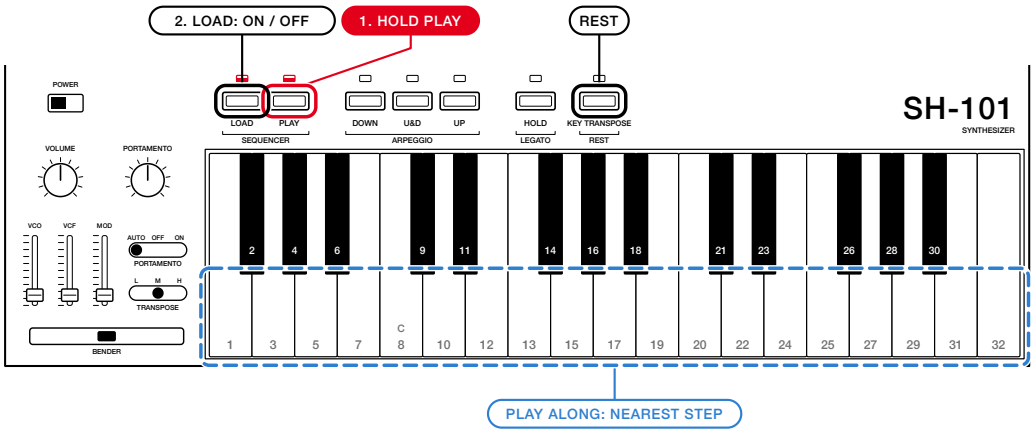
1. While the pattern plays, press LOAD. Playback stops and the LOAD LED lights.
2. The keys overwrite from the step after the one that was sounding. Steps after the ones you correct stay as they were.
3. LOAD again stores.

NOTE

The stock SH-101's Correcting and Adding. LOAD from stop starts a new pattern instead.

13 REAL-TIME RECORD

SEQUENCER PLAY | PLAYING



LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
BLINK	BLINK	OFF	OFF	OFF	OFF	OFF

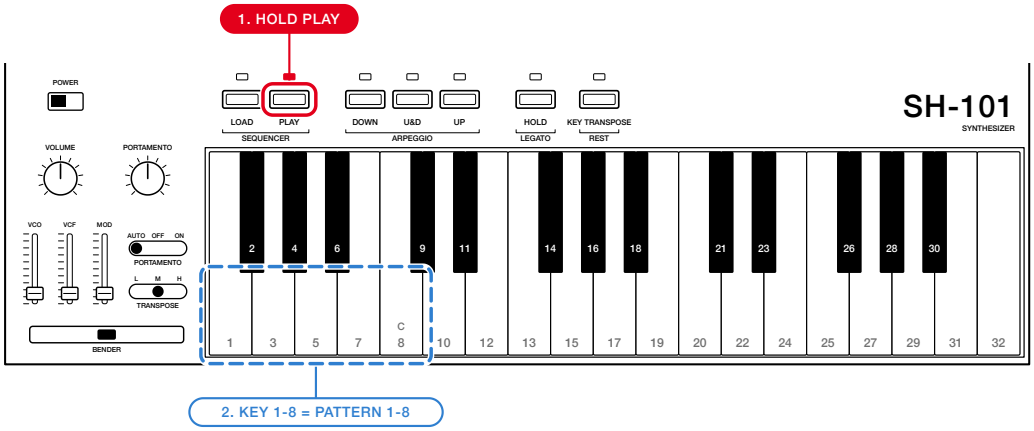
PLAY and LOAD alternating: real-time record.

1. While playing, hold PLAY and press LOAD: the PLAY and LOAD LEDs alternate.
2. Keys overwrite the step they land on (the next step once the step is half over). A key held past the half-step ties into the next step. KEY TRANSPOSE writes a rest.
3. Hold PLAY and press LOAD again to end. The edits are stored when playback stops.

NOTE
An empty pattern gets 16 rest steps to record into.

14 PATTERN SELECT

SEQUENCER | STOPPED OR PLAYING



LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	ON	OFF	OFF	OFF	OFF	OFF

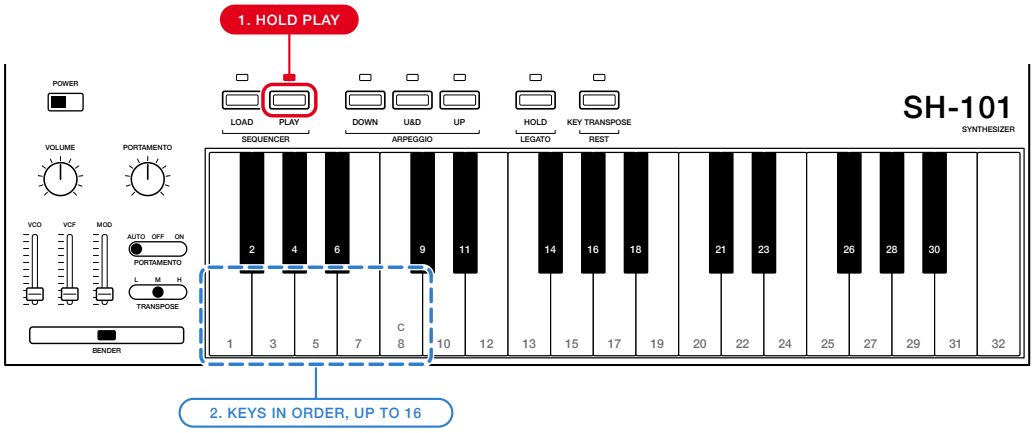
Pattern 2 picked from stop: it plays when PLAY is released.

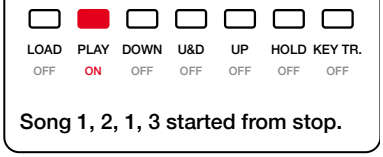
1. Hold PLAY and press one of the 8 lowest keys (key 1 = pattern 1).
2. From stop, the pattern starts when you release PLAY.
3. While playing, it follows at the end of the current pass.

NOTE
MIDI program change 0 to 7 does the same. Hold PLAY and press nothing to toggle play as usual.

15 SONG

SEQUENCER | STOPPED OR PLAYING





Song 1, 2, 1, 3 started from stop.

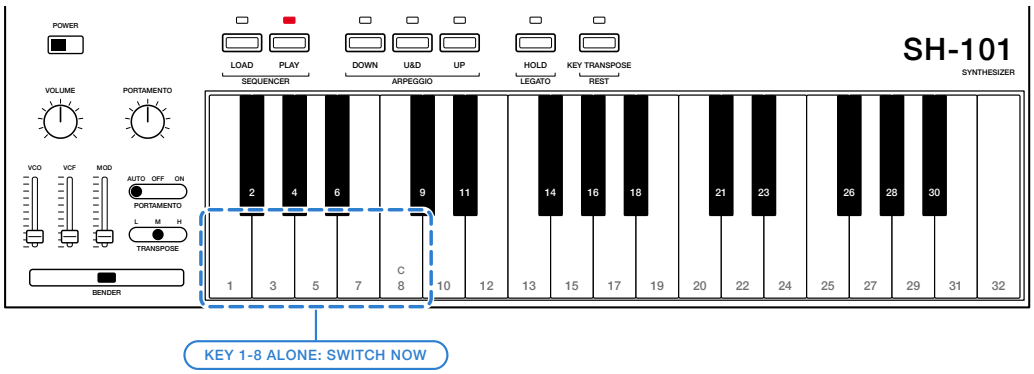
1. Hold PLAY and press several of keys 1-8 in the order to play them.
2. Release PLAY: from stop the song starts; while playing it starts at the end of the pass.
3. Each pattern plays once, then the next; the song loops.

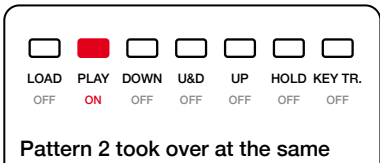
NOTE

Picking a single pattern, or an instant switch, ends the song.

16 INSTANT SWITCH

SEQUENCER PLAY | PLAYING





Pattern 2 took over at the same position.

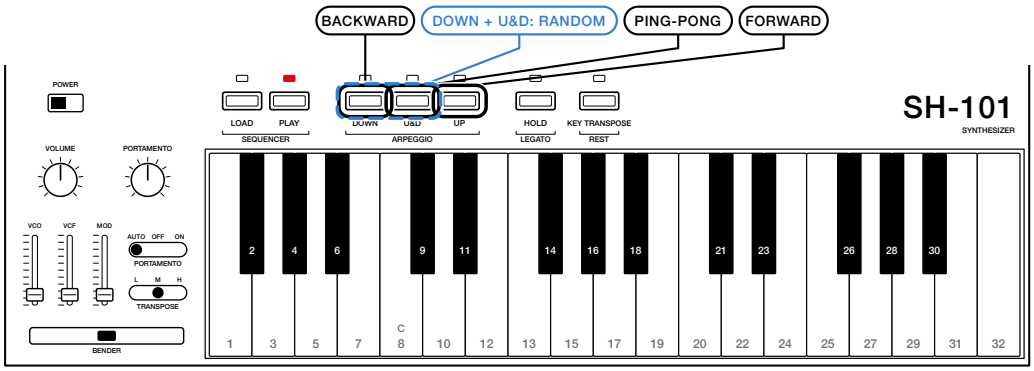
1. While playing, press one of keys 1-8 with nothing else held.
2. That pattern takes over at once, at the same step position (modulo its length).

NOTE

Patterns whose lengths are multiples of each other stay in step.

17 PLAY DIRECTION

SEQUENCER PLAY | PLAYING





PLAY stays lit: the arpeggio LEDs do not show the direction.

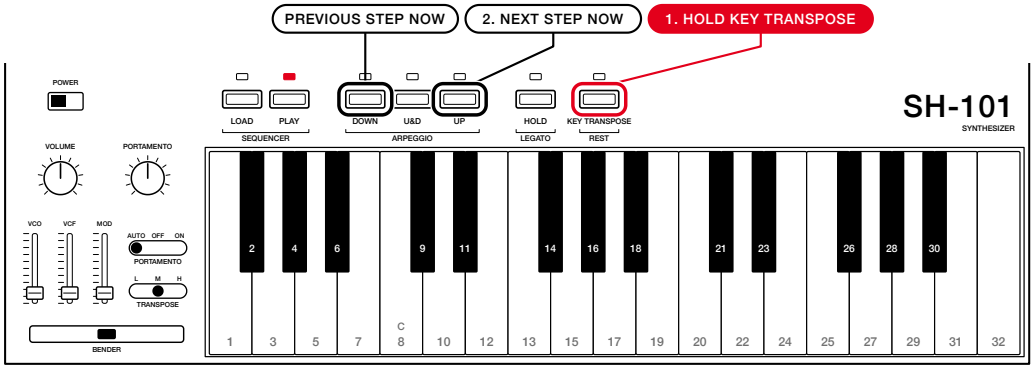
1. While playing, press UP (forward), DOWN (backward), U&D (ping-pong), or DOWN and U&D together (random).
2. The direction is kept across power-off.

NOTE

Config menu keys 15-18 set it too.

18 SHIFT

SEQUENCER PLAY | PLAYING



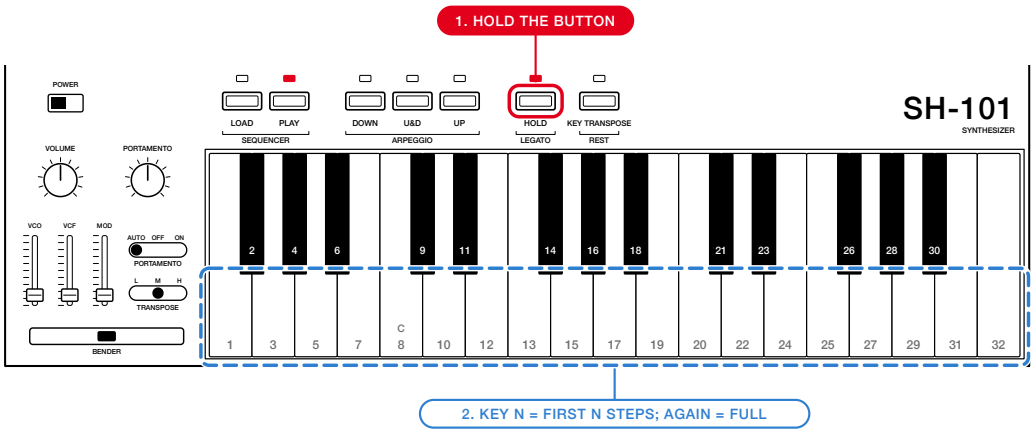


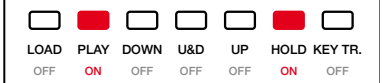
PLAY lit; the pattern moved one step ahead of the clock.

1. While playing, hold KEY TRANSPOSE.
2. UP plays the next step at once, DOWN the previous one: the pattern moves against the clock.

19 LOOP LENGTH, LIVE

SEQUENCER PLAY | PLAYING





Key 2 with HOLD held: only the first 2 steps loop.

1. While playing, hold the HOLD button and press key n: only the first n steps loop (key 1 = 1 step).
2. The same key again: the full pattern.

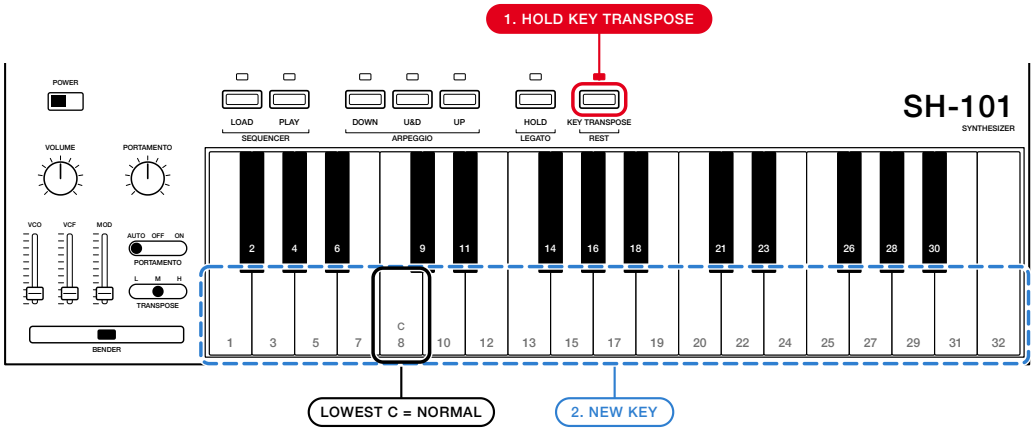
NOTE

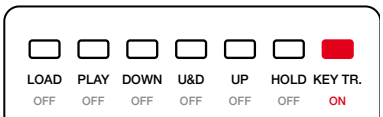
Nothing is erased. The HOLD pedal does not do this; it keeps its hold (or transpose) job.

TRANSPOSE AND TUNE

20 KEY TRANSPOSE

ANY MODE BUT LOAD | ANY TIME





KEY TRANSPOSE lit: transposed up 4 semitones (key 12 = E).

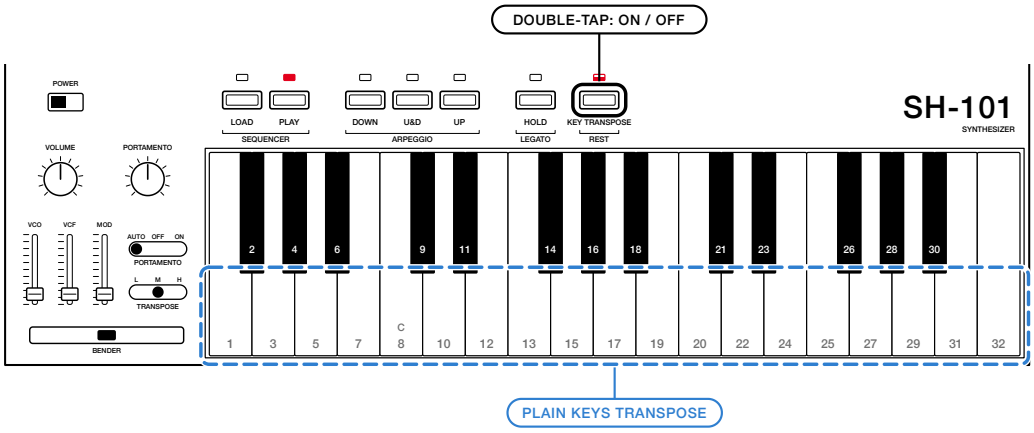
1. Hold KEY TRANSPOSE and press a key: everything that plays moves by its distance from the lowest C (key 8). The LED stays lit.
2. It stays after you let go: keyboard, arpeggio and sequencer all follow.
3. KEY TRANSPOSE + key 8 returns to normal and the LED goes out.

NOTE

Unlike the stock SH-101 the transpose also reaches CV OUT and MIDI OUT.

21 TRANSPOSE MODE

SEQUENCER PLAY OR HELD ARPEGGIO | ANY TIME



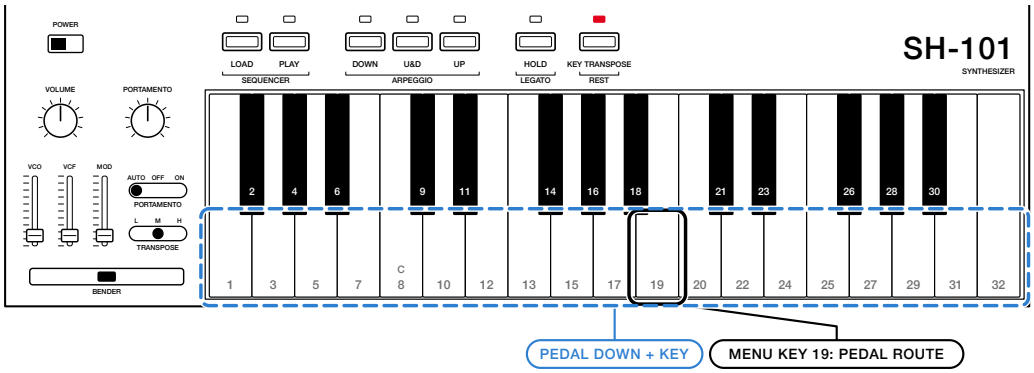
LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	ON	OFF	OFF	OFF	OFF	BLINK

KEY TRANSPOSE blinking: transpose mode. Key 10 moved the pattern up 2.

1. Tap KEY TRANSPOSE twice quickly: its LED blinks.
2. While the sequencer plays, or an arpeggio runs latched by HOLD, plain keys transpose: no button to hold.
3. Double-tap again to leave; the last transpose stays.

22 PEDAL AS KEY TRANSPOSE

CONFIG MENU, THEN ANY MODE | SET ONCE



LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	OFF	OFF	OFF	OFF	OFF	ON

Pedal down + key 12 after key 19: KEY TRANSPOSE lit, transposed.

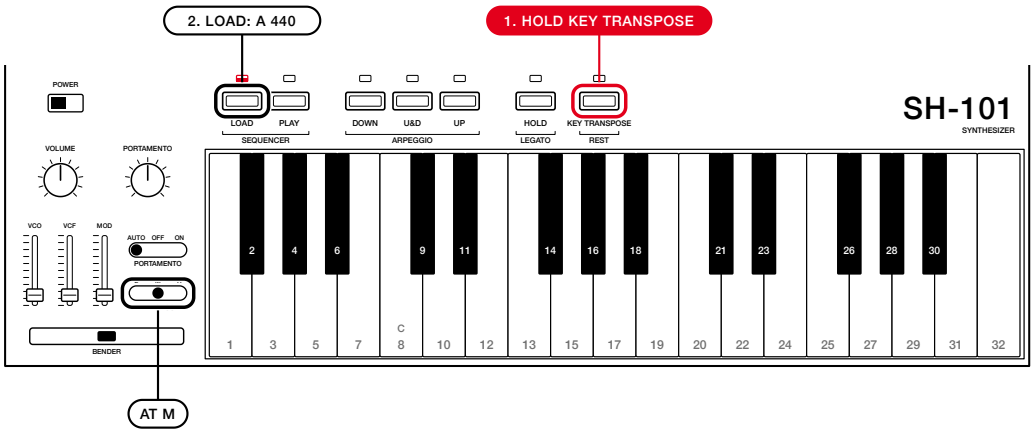
1. In the config menu press key 19: the HOLD pedal jack now acts as the KEY TRANSPOSE button.
2. Hold the pedal and press a key to transpose; double-tap it for transpose mode; in LOAD it enters a rest.
3. Key 19 again in the menu: back to HOLD.

NOTE

Kept across power-off. The HOLD button keeps working as HOLD.

23 TUNE

KEYBOARD | KEYBOARD MODE



LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
BLINK	OFF	OFF	OFF	OFF	OFF	OFF

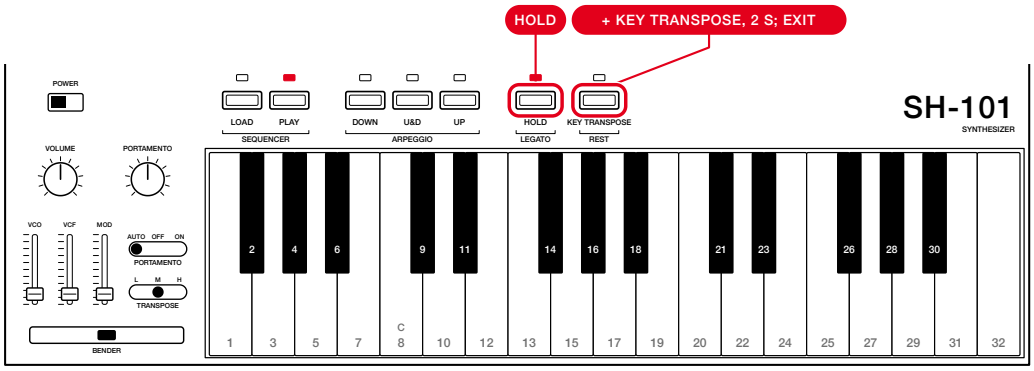
LOAD blinking: the tune note is sounding.

1. From keyboard mode, hold KEY TRANSPOSE and press LOAD: the LOAD LED blinks and an A sounds with the gate open.
2. At 8' with TRANSPOSE on M it is 440 Hz; MIDI OUT sends note 69 for a tuner.
3. Set TUNE, then press any function button to leave.

CONFIG MENU

24 ENTER AND LEAVE

CONFIG MENU | ANY TIME



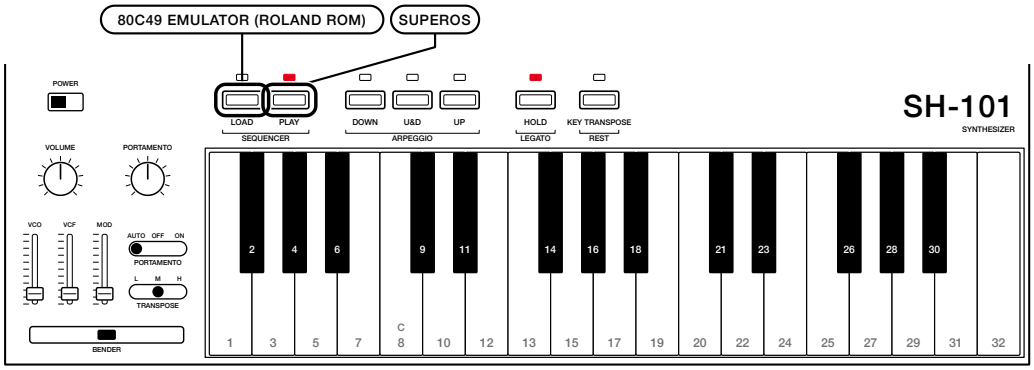
LOAD	PLAY	DOWN	U&D	UP	HOLD	KEY TR.
OFF	ON	OFF	OFF	OFF	ON	OFF

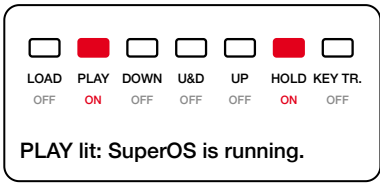
Menu: HOLD lit; PLAY lit = SuperOS is running.

1. Hold HOLD and KEY TRANSPOSE together for 2 seconds (or while switching on).
2. The HOLD LED lights, the gate closes, and LOAD (80C49 emulator) or PLAY (SuperOS) shows the firmware that is running. KEY TRANSPOSE lit = MIDI clock sync on.
3. Press KEY TRANSPOSE to leave. Every setting is kept across power-off.

25 FIRMWARE: SUPEROS OR ORIGINAL

CONFIG MENU | IN THE MENU





PLAY lit: SuperOS is running.

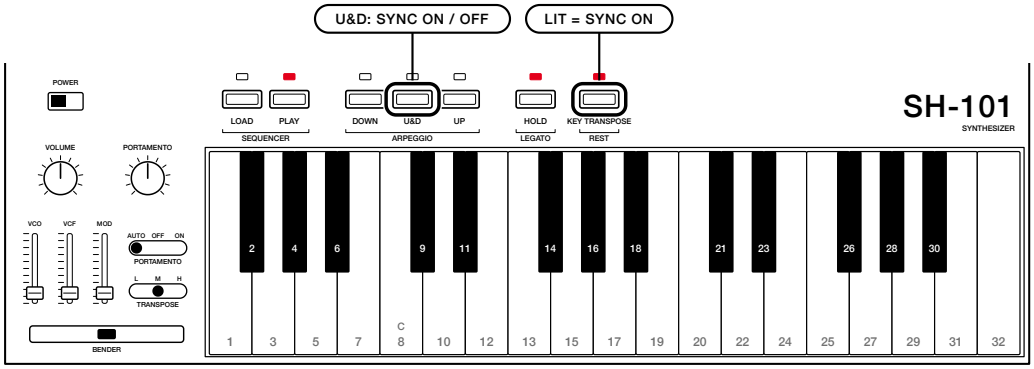
1. In the menu, LOAD restarts into the 80C49 emulator: the original Roland program.
2. PLAY restarts into SuperOS.
3. The choice is kept; the menu works the same way in both.

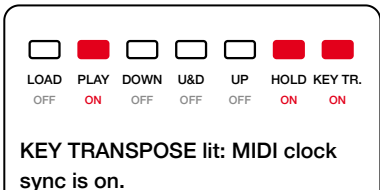
NOTE

In emulator mode the stock SH-101 behaves as from the factory (100 steps, stock arpeggio), with MIDI and clock sync added. Its power-on test mode (LOAD + KEY TRANSPOSE) works there.

26 MIDI CLOCK SYNC

CONFIG MENU | IN THE MENU



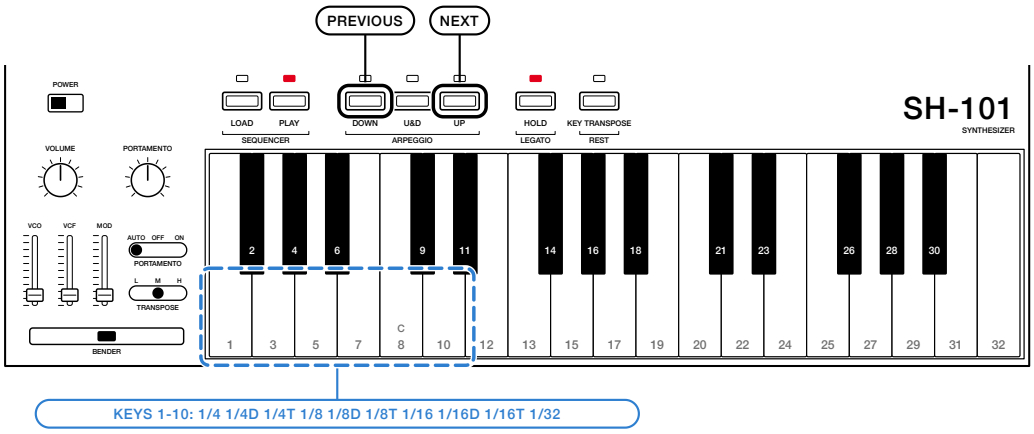


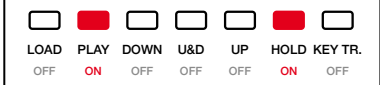
KEY TRANSPOSE lit: MIDI clock sync is on.

1. In the menu, press U&D to switch MIDI clock sync on or off.
2. KEY TRANSPOSE lit = on. The arpeggio and sequencer then follow MIDI clock (MIDI Sync).

27 CLOCK DIVISION

CONFIG MENU | IN THE MENU





Key 7: one step per 1/16 (the default).

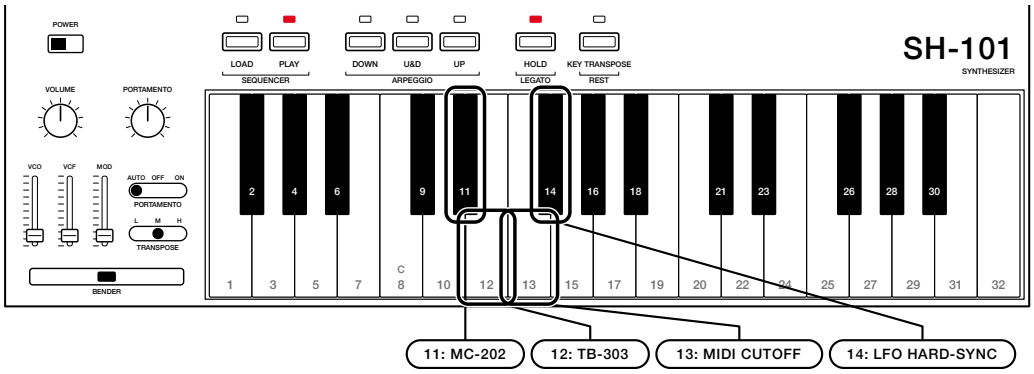
1. In the menu, press one of keys 1-10 for the step length: 1/4, 1/4 dotted, 1/4 triplet, 1/8, 1/8 dotted, 1/8 triplet, 1/16, 1/16 dotted, 1/16 triplet, 1/32.
2. Or step through them with UP and DOWN.

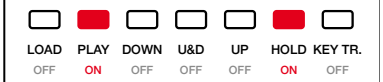
NOTE

No LED shows the division. It applies with MIDI sync on.

28 ACCENT, CUTOFF, LFO SYNC

CONFIG MENU | IN THE MENU





Key 12: TB-303 accent model.

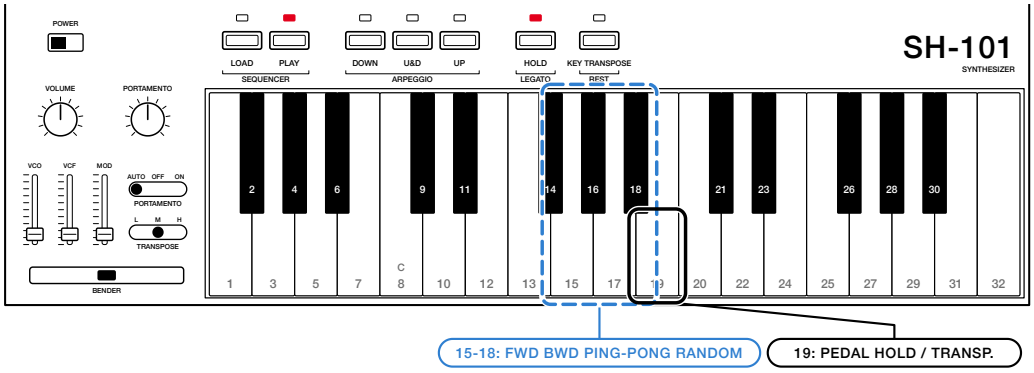
1. Key 11 / 12: accent model, MC-202 or TB-303 style, for notes with a high MIDI velocity (on the filter and the VCA).
2. Key 13: MIDI filter cutoff (CC 74) on / off.
3. Key 14: LFO hard-sync on / off. On: with MIDI sync the LFO is reset to the clock. Off: the LFO runs free for modulation while the steps still follow the clock.

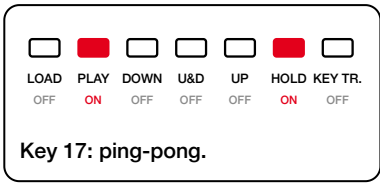
NOTE

Accent and cutoff need the CV DAC (board v1.1 and later). No LED shows these settings.

29 DIRECTION AND PEDAL

CONFIG MENU | IN THE MENU

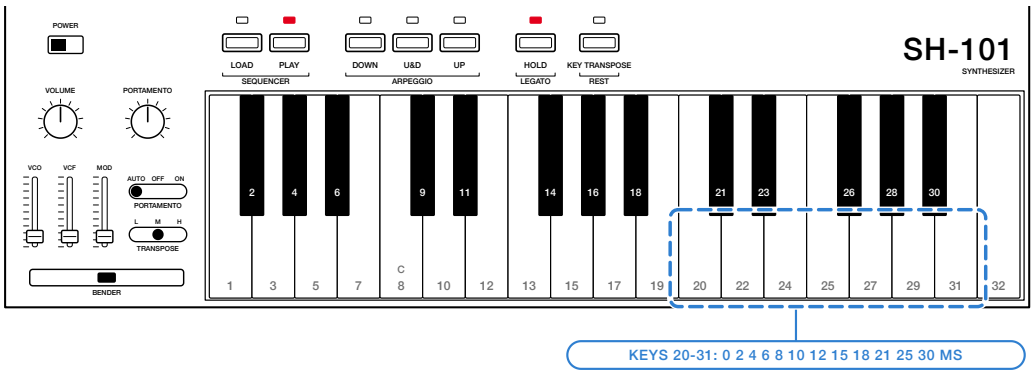


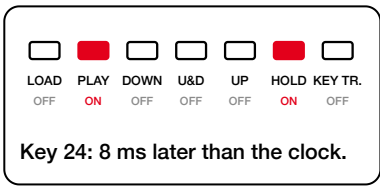


1. Keys 15-18 set the sequencer direction: forward, backward, ping-pong, random (the same setting as the buttons while playing).
2. Key 19 switches the pedal jack between HOLD and KEY TRANSPOSE.

30 SYNC DELAY

CONFIG MENU | IN THE MENU





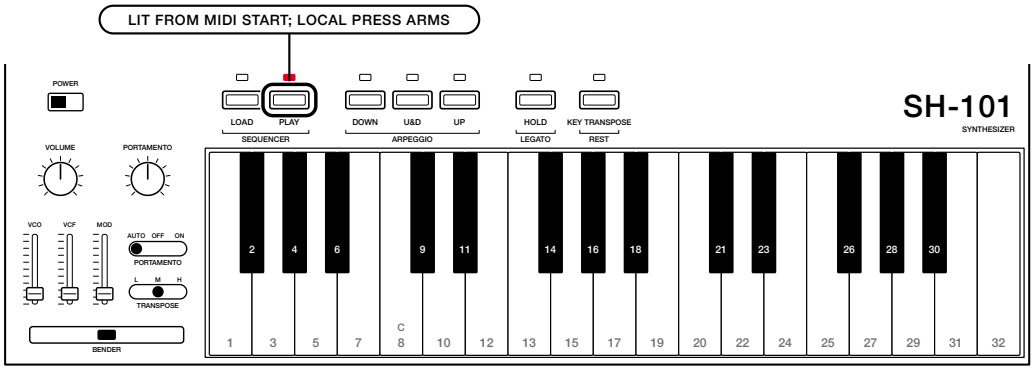
1. Keys 20-31 set how much later than the incoming MIDI clock the SH-101 plays: 0, 2, 4, 6, 8, 10, 12, 15, 18, 21, 25 or 30 ms.
2. Use it to line the SH-101 up with slower gear or a delayed listening path.

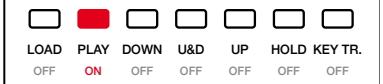
NOTE

Only with MIDI sync on. The editor can set 0 to 60 ms.

31 HOST TRANSPORT

SEQUENCER, SYNC ON | MIDI CLOCK SYNC ON





PLAY lit: the host's Start started the sequencer.

1. With sync on, MIDI Start starts the sequencer from the first clock, MIDI Stop stops it.
2. Pressing PLAY yourself only arms it: the first step waits for the host's clock.
3. A synced SH-101 never runs on its own: no clock, no steps. The arpeggio follows the clock too.

NOTE

Start does not interrupt a running arpeggio or LOAD. With sync off, Start only rewinds a pattern that is already playing.

LEDS

LED	MEANING
LOAD lit	Recording (LOAD); in the menu: 80C49 emulator running
LOAD blinks	Tune
PLAY lit	Sequencer playing (or armed); in the menu: SuperOS running
PLAY + LOAD alternate	Real-time record
DOWN / U&D / UP lit	Arpeggio direction
U&D blinks	Random arpeggio
HOLD lit	HOLD button or pedal down; in the menu: the menu is open
KEY TRANSPOSE lit	A transpose is active; in the menu: MIDI clock sync on
KEY TRANSPOSE blinks	Transpose mode

PLAYING

GESTURE	FUNCTION
Keys	play; the lowest held key sounds
HOLD button or pedal down + keys	keys latch while it is down
TRANSPOSE switch L / M / H	octave down / normal / up
PORTAMENTO switch AUTO	only legato notes glide

ARPEGGIO

GESTURE	FUNCTION
UP / DOWN / U&D	arpeggio; same button again = off
DOWN + U&D together	random arpeggio; again = off
Arp on: KEY TRANSPOSE + UP / DOWN	octave range 1-3
Arp + HOLD or pedal + chord	latched arpeggio

SEQUENCER

GESTURE	FUNCTION
Stopped: LOAD, keys, LOAD	record a new pattern
In LOAD: HOLD + key / KEY TRANSPOSE / UP / DOWN	legato / rest / ratchet / undo
PLAY	play / stop
Playing: LOAD	re-record from the next step
Playing: PLAY + LOAD	real-time record on / off
PLAY + key 1-8	pattern (stopped: starts; playing: at loop end)
PLAY + several keys 1-8	song, up to 16, loops
Playing: key 1-8 alone	switch pattern now
Playing: UP / DOWN / U&D / DOWN + U&D	forward / backward / ping-pong / random
Playing: KEY TRANSPOSE + UP / DOWN	shift a step ahead / back
Playing: HOLD button + key n	loop the first n steps; again = full

TRANSPOSE AND TUNE

GESTURE	FUNCTION
KEY TRANSPOSE + key	transpose; + key 8 (lowest C) = normal
Double-tap KEY TRANSPOSE	transpose mode on / off
Menu key 19	pedal acts as KEY TRANSPOSE
Keyboard mode: KEY TRANSPOSE + LOAD	tune note A 440; any button exits

CONFIG MENU

GESTURE	FUNCTION
HOLD + KEY TRANSPOSE 2 s	config menu; KEY TRANSPOSE exits
Menu: LOAD / PLAY	80C49 emulator / SuperOS
Menu: U&D	MIDI clock sync on / off
Menu: keys 1-10, UP / DOWN	clock division
Menu: keys 11-14	accent MC-202 / TB-303, MIDI cutoff, LFO hard-sync
Menu: keys 15-18 / 19	sequencer direction / pedal route
Menu: keys 20-31	sync delay 0-30 ms

MIDI SYNC

GESTURE	FUNCTION
Sync on: MIDI Start / Stop	run the sequencer; PLAY only arms

MIDI

Notes 53 to 84 on the receive channel act as keys 1 to 32 (they arpeggiate and record); other notes 36 to 99 play the voice. Program change 0 to 7 selects pattern 1 to 8. Everything played goes out on MIDI OUT. The web editor (tools/web-editor) edits patterns and settings over USB-C or DIN.

ABOUT THESE PAGES

The LEDs in every drawing are what the SuperOS-101 firmware (rev 135021e*) shows in the host simulator (sim/superos101-sim, the real firmware source) after the keys of that function, on a fresh machine. The panel drawing follows the Roland owner's manual; most drawings show the button strip, controllers and keyboard. Details: README.md.